

Fundamentals of Capital Budgeting

Solutions

- 8-2. Kokomochi is considering the launch of an advertising campaign for its latest dessert product, the Mini Mochi Munch. Kokomochi plans to spend \$5 million on TV, radio, and print advertising this year for the campaign. The ads are expected to boost sales of the Mini Mochi Munch by \$9 million this year and by \$7 million next year. In addition, the company expects that new consumers who try the Mini Mochi Munch will be more likely to try Kokomochi's other products. As a result, sales of other products are expected to rise by \$2 million each year.

Kokomochi's gross profit margin for the Mini Mochi Munch is 35%, and its gross profit margin averages 25% for all other products. The company's marginal corporate tax rate is 35% both this year and next year. What are the incremental earnings associated with the advertising campaign?

	A	B	C	D	E
1			Year	1	2
2	Incremental Earnings Forecast (\$000s)				
3	1	Sales of Mini Mochi Munch		9,000	7,000
4	2	Other Sales		2,000	2,000
5	3	Cost of Goods Sold		(7,350)	(6,050)
6	4	Gross Profit		3,650	2,950
7	5	Selling, General & Admin.		(5,000)	-
8	6	Depreciation		-	-
9	7	EBIT		(1,350)	2,950
10	8	Income tax at 35%		473	(1,033)
11	9	Unlevered Net Income		(878)	1,918

- 8-3. Home Builder Supply, a retailer in the home improvement industry, currently operates seven retail outlets in Georgia and South Carolina. Management is contemplating building an eighth retail store across town from its most successful retail outlet. The company already owns the land for this store, which currently has an abandoned warehouse located on it. Last month, the marketing department spent \$10,000 on market research to determine the extent of customer demand for the new store. Now Home Builder Supply must decide whether to build and open the new store.

Which of the following should be included as part of the incremental earnings for the proposed new retail store?

- The cost of the land where the store will be located.
 - The cost of demolishing the abandoned warehouse and clearing the lot.
 - The loss of sales in the existing retail outlet, if customers who previously drove across town to shop at the existing outlet become customers of the new store instead.
 - The \$10,000 in market research spent to evaluate customer demand.
 - Construction costs for the new store.
 - The value of the land if sold.
 - Interest expense on the debt borrowed to pay the construction costs.
- No, this is a sunk cost and will not be included directly. (But see (f) below.)
 - Yes, this is a cost of opening the new store.

- c. Yes, this loss of sales at the existing store should be deducted from the sales at the new store to determine the incremental increase in sales that opening the new store will generate for HBS.
- d. No, this is a sunk cost.
- e. This is a capital expenditure associated with opening the new store. These costs will, therefore, increase HBS's depreciation expenses.
- f. Yes, this is an opportunity cost of opening the new store. (By opening the new store, HBS forgoes the after-tax proceeds it could have earned by selling the property. This loss is equal to the sale price less the taxes owed on the capital gain from the sale, which is the difference between the sale price and the book value of the property. The book value equals the initial cost of the property less accumulated depreciation.)
- g. While these financing costs will affect HBS's actual earnings, for capital budgeting purposes we calculate the incremental earnings without including financing costs to determine the project's unlevered net income.

8-4. Hyperion, Inc. currently sells its latest high-speed color printer, the Hyper 500, for \$350. It plans to lower the price to \$300 next year. Its cost of goods sold for the Hyper 500 is \$200 per unit, and this year's sales are expected to be 20,000 units.

- a. Suppose that if Hyperion drops the price to \$300 immediately, it can increase this year's sales by 25% to 25,000 units. What would be the incremental impact on this year's EBIT of such a price drop?
- b. Suppose that for each printer sold, Hyperion expects additional sales of \$75 per year on ink cartridges for the next three years, and Hyperion has a gross profit margin of 70% on ink cartridges. What is the incremental impact on EBIT for the next three years of a price drop this year?

a. Change in EBIT = Gross profit with price drop – Gross profit without price drop

$$= 25,000 \times (300 - 200) - 20,000 \times (350 - 200)$$

$$= -\$500,000$$

b. Change in EBIT from ink cartridge sales = $25,000 \times \$75 \times 0.70 - 20,000 \times \$75 \times 0.70 = \$262,500$

Therefore, incremental change in EBIT for the next three years is

Year 1: $\$262,500 - 500,000 = -\$237,500$

Year 2: $\$262,500$

Year 3: $\$262,500$

8-6. Cellular Access, Inc. is a cellular telephone service provider that reported net income of \$250 million for the most recent fiscal year. The firm had depreciation expenses of \$100 million, capital expenditures of \$200 million, and no interest expenses. Working capital increased by \$10 million. Calculate the free cash flow for Cellular Access for the most recent fiscal year.

$$\text{FCF} = \text{Unlevered Net Income} + \text{Depreciation} - \text{CapEx} - \text{Increase in NWC} = 250 + 100 - 200 - 10 = \$140 \text{ million.}$$

8-7. Castle View Games would like to invest in a division to develop software for video games. To evaluate this decision, the firm first attempts to project the working capital needs for this operation. Its chief financial officer has developed the following estimates (in millions of dollars):

	Year 1	Year 2	Year 3	Year 4	Year 5
Cash	6	12	15	15	15
Accounts Receivable	21	22	24	24	24
Inventory	5	7	10	12	13
Accounts Payable	18	22	24	25	30

Assuming that Castle View currently does not have any working capital invested in this division, calculate the cash flows associated with changes in working capital for the first five years of this investment.

	Year0	Year1	Year2	Year3	Year4	Year5
1 Cash		6	12	15	15	15
2 Accounts Receivable		21	22	24	24	24
3 Inventory		5	7	10	12	13
4 Accounts Payable		18	22	24	25	30
5 Net working capital (1+2+3-4)	0	14	19	25	26	22
6 Increase in NWC		14	5	6	1	-4

- 8-8. Mersey Chemicals manufactures polypropylene that it ships to its customers via tank car. Currently, it plans to add two additional tank cars to its fleet four years from now. However, a proposed plant expansion will require Mersey's transport division to add these two additional tank cars in two years' time rather than in four years. The current cost of a tank car is \$2 million, and this cost is expected to remain constant. Also, while tank cars will last indefinitely, they will be depreciated straight-line over a five-year life for tax purposes. Suppose Mersey's tax rate is 40%. When evaluating the proposed expansion, what incremental free cash flows should be included to account for the need to accelerate the purchase of the tank cars?

Year:	0	1	2	3	4	5	6	7	8	9	10
with expansion											
CapEx			-4								
Depreciation Tax Shield				0.32	0.32	0.32	0.32	0.32			
FCF	0	0	-4	0.32	0.32	0.32	0.32	0.32	0	0	0
without expansion											
CapEx					-4						
Depreciation Tax Shield						0.32	0.32	0.32	0.32	0.32	
FCF	0	0	0	0	-4	0.32	0.32	0.32	0.32	0.32	0
Incremental FCF (with-without)	0	0	-4	0.32	4.32	0	0	0	-0.32	-0.32	0

- 8-9. Elmdale Enterprises is deciding whether to expand its production facilities. Although long-term cash flows are difficult to estimate, management has projected the following cash flows for the first two years (in millions of dollars):

	Year 1	Year 2
Revenues	125	160
Costs of goods sold and operating expenses other than depreciation	40	60
Depreciation	25	36
Increase in net working capital	5	8
Capital expenditures	30	40
Marginal corporate tax rate	35%	35%

- a. What are the incremental earnings for this project for years 1 and 2?
- b. What are the free cash flows for this project for the first two years?

a.

		Year 1	2
Incremental Earnings Forecast (\$000s)			
1	Sales	125.0	160.0
2	Costs of good sold and operating expenses other than depreciation	(40.0)	(60.0)
3	Depreciation	(25.0)	(36.0)
4	EBIT	60.0	64.0
5	Income tax at 35%	(21.0)	(22.4)
6	Unlevered Net Income	39.0	41.6

b.

Free Cash Flow (\$000s)		1	2
7	Plus: Depreciation	25.0	36.0
8	Less: Capital Expenditures	(30.0)	(40.0)
9	Less: Increases in NWC	(5.0)	(8.0)
10	Free Cash Flow	29.0	29.6

- 8-10. You are a manager at Percolated Fiber, which is considering expanding its operations in synthetic fiber manufacturing. Your boss comes into your office, drops a consultant's report on your desk, and complains, "We owe these consultants \$1 million for this report, and I am not sure their analysis makes sense. Before we spend the \$25 million on new equipment needed for this project, look it over and give me your opinion." You open the report and find the following estimates (in thousands of dollars):

	Project Year				
	1	2	...	9	10
Sales revenue	30,000	30,000		30,000	30,000
– Cost of goods sold	18,000	18,000		18,000	18,000
= Gross profit	12,000	12,000		12,000	12,000
– General, sales, and administrative expenses	2,000	2,000		2,000	2,000
– Depreciation	2,500	2,500		2,500	2,500
= Net operating income	7,500	7,500		7,500	7,500
– Income tax	2,625	2,625		2,625	2,625
Net Income	4,875	4,875		4,875	4,875

All of the estimates in the report seem correct. You note that the consultants used straight-line depreciation for the new equipment that will be purchased today (year 0), which is what the accounting department recommended. The report concludes that because the project will increase earnings by \$4.875 million per year for 10 years, the project is worth \$48.75 million. You think back to your halcyon days in finance class and realize there is more work to be done!

First, you note that the consultants have not factored in the fact that the project will require \$10 million in working capital upfront (year 0), which will be fully recovered in year 10. Next, you see they have attributed \$2 million of selling, general and administrative expenses to the project, but you know that \$1 million of this amount is overhead that will be incurred even if the project is not accepted. Finally, you know that accounting earnings are not the right thing to focus on!

- a. Given the available information, what are the free cash flows in years 0 through 10 that should be used to evaluate the proposed project?
- b. If the cost of capital for this project is 14%, what is your estimate of the value of the new project?

a. Free Cash Flows are:

	0	1	2	...	9	10
= Net income		4,875	4,875		4,875	4,875
+ Overhead (after tax at 35%)		650	650		650	650
+ Depreciation		2,500	2,500		2,500	2,500
– Capex	25,000					
– Inc. in NWC	10,000					–10,000
FCF	–35,000	8,025	8,025	...	8,025	18,025

$$b. \quad NPV = -35 + 8.025 \times \frac{1}{.14} \left(1 - \frac{1}{1.14^9} \right) + \frac{18.025}{1.14^{10}} = 9.56$$

- 8-19. Bay Properties is considering starting a commercial real estate division. It has prepared the following four-year forecast of free cash flows for this division:

	Year 1	Year 2	Year 3	Year 4
Free Cash Flow	–\$185,000	\$12,000	\$99,000	\$240,000

Assume cash flows after year 4 will grow at 3% per year, forever. If the cost of capital for this division is 14%, what is the continuation value in year 4 for cash flows after year 4? What is the value today of this division?

The expected cash flow in year 5 is $240,000 \times 1.03 = 247,200$. We can value the cash flows in year 5 and beyond as a growing perpetuity:

$$\text{Continuation Value in Year 4} = 247,200 / (0.14 - 0.03) = \$2,247,273$$

We can then compute the value of the division by discounting the FCF in years 1 through 4, together with the continuation value:

$$NPV = \frac{-185,000}{1.14} + \frac{12,000}{1.14^2} + \frac{99,000}{1.14^3} + \frac{240,000 + 2,247,273}{1.14^4} = \$1,386,440$$

- 8-20. Your firm would like to evaluate a proposed new operating division. You have forecasted cash flows for this division for the next five years, and have estimated that the cost of capital is 12%. You would like to estimate a continuation value. You have made the following forecasts for the last year of your five-year forecasting horizon (in millions of dollars):

	Year 5
Revenues	1200
Operating income	100
Net income	50
Free cash flows	110
Book value of equity	400

- a. You forecast that future free cash flows after year 5 will grow at 2% per year, forever. Estimate the continuation value in year 5, using the perpetuity with growth formula.
- b. You have identified several firms in the same industry as your operating division. The average P/E ratio for these firms is 30. Estimate the continuation value assuming the P/E

ratio for your division in year 5 will be the same as the average P/E ratio for the comparable firms today.

- c. The average market/book ratio for the comparable firms is 4.0. Estimate the continuation value using the market/book ratio.

a. $FCF \text{ in year } 6 = 110 \times 1.02 = 112.2$

Continuation Value in year 5 = $112.2 / (12\% - 2\%) = \$1,122$.

- b. We can estimate the continuation value as follows:

Continuation Value in year 5 = (Earnings in year 5) $\times$ (P/E ratio in year 5)
 $= \$50 \times 30 = \1500 .

- c. We can estimate the continuation value as follows:

Continuation Value in year 5 = (Book value in year 5) $\times$ (M/B ratio in year 5)
 $= \$400 \times 4 = \1600 .

- 8-24. **Bauer Industries is an automobile manufacturer. Management is currently evaluating a proposal to build a plant that will manufacture lightweight trucks. Bauer plans to use a cost of capital of 12% to evaluate this project. Based on extensive research, it has prepared the following incremental free cash flow projections (in millions of dollars):**

	Year 0	Years 1–9	Year 10
Revenues		100.0	100.0
– Manufacturing expenses (other than depreciation)		–35.0	–35.0
– Marketing expenses		–10.0	–10.0
– Depreciation		–15.0	–15.0
= EBIT		40.0	40.0
– Taxes (35%)		–14.0	–14.0
= Unlevered net income		26.0	26.0
+ Depreciation		+15.0	+15.0
– Increases in net working capital		–5.0	–5.0
– Capital expenditures	–150.0		
+ Continuation value			+12.0
= Free cash flow	–150.0	36.0	48.0

- a. For this base-case scenario, what is the NPV of the plant to manufacture lightweight trucks?
- b. Based on input from the marketing department, Bauer is uncertain about its revenue forecast. In particular, management would like to examine the sensitivity of the NPV to the revenue assumptions. What is the NPV of this project if revenues are 10% higher than forecast? What is the NPV if revenues are 10% lower than forecast?
- c. Rather than assuming that cash flows for this project are constant, management would like to explore the sensitivity of its analysis to possible growth in revenues and operating expenses. Specifically, management would like to assume that revenues, manufacturing expenses, and marketing expenses are as given in the table for year 1 and grow by 2% per year every year starting in year 2. Management also plans to assume that the initial capital expenditures (and therefore depreciation), additions to working capital, and continuation value remain as initially specified in the table. What is the NPV of this project under these alternative assumptions? How does the NPV change if the revenues and operating expenses grow by 5% per year rather than by 2%?

- d. To examine the sensitivity of this project to the discount rate, management would like to compute the NPV for different discount rates. Create a graph, with the discount rate on the x-axis and the NPV on the y-axis, for discount rates ranging from 5% to 30%. For what ranges of discount rates does the project have a positive NPV?

	Year	0	1	2	3	4	5	6	7	8	9	10
Free Cash Flow Forecast (\$ millions)												
1	Sales	—	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
2	Manufacturing	—	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)	(35.0)
3	Marketing Expenses	—	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)
4	Depreciation	—	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)	(15.0)
5	EBIT	—	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
6	Income tax at 35%	—	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)	(14.0)
7	Unlevered Net Income	—	26.0	26.0	26.0	26.0	26.0	26.0	26.0	26.0	26.0	26.0
8	Depreciation	—	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
9	Inc. in NWC	—	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
10	Capital Expenditures	(150.0)	—	—	—	—	—	—	—	—	—	—
11	Continuation value	—	—	—	—	—	—	—	—	—	—	12.0
12	Free Cash Flow	(150.0)	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	48.0
13	NPV at 12%	57.3	—	—	—	—	—	—	—	—	—	—

- a. The NPV of the estimate free cash flow is

$$\text{NPV} = -150 + 36 \times \frac{1}{0.12} \left(1 - \frac{1}{1.12^9} \right) + \frac{48}{1.12^{10}} = \$57.3 \text{ million.}$$

- b. Initial Sales 90 100 110
 NPV 20.5 57.3 94.0

- c. Growth Rate 0% 2% 5%
 NPV 57.3 72.5 98.1

- d. NPV is positive for discount rates below the IRR of 20.6%.

